

1. Derive the spectral density of the following stationary stochastic processes. $X_t = t + W_t$, where W_t is an iid white noise process with $Var(W_t) = \sigma^2$. For each process, consider that W_t is an iid white noise process with $Var(W_t) = \sigma^2$.

(a) $X_t = 5 + W_t$.

Solution: The auto-covariance was derived in Foundations II Exercises:

$$\gamma_h = \begin{cases} \sigma^2 & \text{if } h = 0 \\ 0 & \text{otherwise} \end{cases}$$

The spectral density is the Fourier transform of the auto-covariance function. Since the process is white noise with a mean of 5 and no correlation between different time points ($\gamma_h = 0$ for $h \neq 0$), the spectral density is constant across all frequencies:

$$f(\omega) = \sum_{h=-\infty}^{\infty} \gamma_h e^{-2\pi i \omega h}.$$

Substituting the auto-covariance:

$$f(\omega) = \gamma_0 = \sigma^2.$$

Thus, the spectral density is:

$$f(\omega) = \sigma^2.$$

(b) $X_t = 2W_t + W_{t-1}$.

Solution: The auto-covariance was derived in Foundations II Exercises:

$$\gamma_h = \begin{cases} 5\sigma^2 & \text{if } h = 0 \\ 2\sigma^2 & \text{if } h = \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

The spectral density is:

$$f(\omega) = \sum_{h=-\infty}^{\infty} \gamma_h e^{-2\pi i \omega h}.$$

Substituting the auto-covariance:

$$f(\omega) = 5\sigma^2 + 2\sigma^2 (e^{-2\pi i \omega} + e^{2\pi i \omega}).$$

Using the property $e^{ix} + e^{-ix} = 2\cos(x)$, we simplify:

$$f(\omega) = 5\sigma^2 + 4\sigma^2 \cos(2\pi\omega).$$

Thus, the spectral density is:

$$f(\omega) = \sigma^2 (5 + 4\cos(2\pi\omega)).$$

(c) $X_t = (-1)^t W_t$

Solution: The auto-covariance was derived in Foundations II Exercises:

$$\gamma_h = \begin{cases} \sigma^2 & \text{if } h = 0 \\ 0 & \text{otherwise} \end{cases}$$

The spectral density is:

$$f(\omega) = \sum_{h=-\infty}^{\infty} \gamma_h e^{-2\pi i \omega h} = \sigma^2.$$

(d) $X_t = W_t - \frac{1}{2}W_{t-2}.$

Solution: The auto-covariance can be expanded as follows:

$$\begin{aligned} \text{cov}(X_{t+h}, X_t) &= \text{cov}(W_{t+h} - \frac{1}{2}W_{t+h-2}, W_t - \frac{1}{2}W_{t-2}) \\ &= \text{cov}(W_{t+h}, W_t) - \frac{1}{2}\text{cov}(W_{t+h}, W_{t-2}) \\ &\quad - \frac{1}{2}\text{cov}(W_{t+h-2}, W_t) + \frac{1}{4}\text{cov}(W_{t+h-2}, W_{t-2}) \\ &= \begin{cases} \frac{5}{4}\sigma^2 & \text{if } h = 0, \\ -\frac{1}{2}\sigma^2 & \text{if } h = \pm 2, \\ 0 & \text{otherwise.} \end{cases} \\ &= \gamma(h) \end{aligned}$$

The spectral density is:

$$\begin{aligned} f(\omega) &= \sum_{h=-\infty}^{\infty} \gamma_h e^{-2\pi i \omega h} \\ &= \gamma_0 + \gamma_2 e^{-2\pi i \omega 2} + \gamma_{-2} e^{2\pi i \omega 2} \\ &= \frac{5}{4}\sigma^2 - \frac{1}{2}\sigma^2 (e^{-i4\pi\omega} + e^{i4\pi\omega}) \\ &= \frac{5}{4}\sigma^2 - \frac{1}{2}\sigma^2 \cdot 2 \cos(4\pi\omega) \\ &= \sigma^2 \left(\frac{5}{4} - \cos(4\pi\omega) \right). \end{aligned}$$

2. Derive the auto-covariance from the spectral density $f(\omega) = 1 + \cos(2\pi\omega)$.

For this derivation, it is useful to note the following result for the integral of a complex exponential:

$$\int_{-1/2}^{1/2} e^{2\pi i \omega k} d\omega = \begin{cases} 1 & \text{if } k = 0, \\ \frac{\sin(\pi k)}{\pi k} & \text{if } k \neq 0. \end{cases}$$

Solution: The auto-covariance function is the inverse Fourier transform of the spectral density:

$$\gamma(h) = \int_{-1/2}^{1/2} f(\omega) e^{2\pi i \omega h} d\omega = \int_{-1/2}^{1/2} (1 + \cos(2\pi\omega)) e^{2\pi i \omega h} d\omega.$$

Split the integral:

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i \omega h} d\omega + \int_{-1/2}^{1/2} \cos(2\pi\omega) e^{2\pi i \omega h} d\omega.$$

Using the provided result and the observation that $\sin(\pi k) = 0$ for integer k , the first integral evaluates to 1 for $h = 0$ and 0 otherwise.

Using Euler's identity $\cos(2\pi\omega) = \frac{e^{2\pi i \omega} + e^{-2\pi i \omega}}{2}$, we rewrite the second integral as

$$\int_{-1/2}^{1/2} \frac{e^{2\pi i \omega} + e^{-2\pi i \omega}}{2} e^{2\pi i \omega h} d\omega.$$

Split the integral:

$$\frac{1}{2} \left(\int_{-1/2}^{1/2} e^{2\pi i \omega (h+1)} d\omega + \int_{-1/2}^{1/2} e^{2\pi i \omega (h-1)} d\omega \right).$$

Evaluate each term using the provided result:

$$\int_{-1/2}^{1/2} e^{2\pi i \omega (h+1)} d\omega = \begin{cases} 1 & \text{if } h = -1, \\ \frac{\sin(\pi(h+1))}{\pi(h+1)} & \text{if } h \neq -1, \end{cases}$$

and

$$\int_{-1/2}^{1/2} e^{2\pi i \omega (h-1)} d\omega = \begin{cases} 1 & \text{if } h = 1, \\ \frac{\sin(\pi(h-1))}{\pi(h-1)} & \text{if } h \neq 1, \end{cases}$$

Since $\sin(\pi k) = 0$ for integer k , the auto-covariance function is:

$$\gamma(h) = \begin{cases} 1 & \text{if } h = 0, \\ \frac{1}{2} & \text{if } h = \pm 1, \\ 0 & \text{otherwise.} \end{cases}$$

3. Derive the mean, variance, auto-covariance, and spectral density of a mixture of periodic series:

$$X_t = \sum_{j=1}^k (A_j c_{t,j} + B_j s_{t,j}),$$

where $A_j, B_j \sim \mathcal{N}(0, \sigma_j^2)$ are uncorrelated $\forall j$, $c_{t,j} = \cos(2\pi\omega_j t)$, and $s_{t,j} = \sin(2\pi\omega_j t)$.

Solution:

- $E[X_t] = \sum_{j=1}^k (c_{t,j}E[A_j] + s_{t,j}E[B_j]) = 0$ since $E[A_j] = E[B_j] = 0 \forall j$, and $c_{t,j}$ and $s_{t,j}$ are deterministic functions.
- $Var(X_t) = \sum_{j=1}^k (c_{t,j}^2 Var(A_j) + s_{t,j}^2 Var(B_j)) = \sum_{j=1}^k \sigma_j^2 (c_{t,j}^2 + s_{t,j}^2) = \sum_{j=1}^k \sigma_j^2$ since A_j and B_j are uncorrelated for all j , and $c_{t,j}^2 + s_{t,j}^2 = 1 \forall j, t$.
- The auto-covariance is given by:

$$\begin{aligned} cov(X_{t+h}, X_t) &= cov\left(\sum_{j=1}^k (A_j c_{t+h,j} + B_j s_{t+h,j}), \sum_{l=1}^k (A_l c_{t,l} + B_l s_{t,l})\right) \\ &= \sum_{j=1}^k \sum_{l=1}^k cov(A_j c_{t+h,j} + B_j s_{t+h,j}, A_l c_{t,l} + B_l s_{t,l}). \end{aligned}$$

Since $A_j, B_j \sim \mathcal{N}(0, \sigma_j^2)$ are uncorrelated $\forall j$, this expression reduces to:

$$\begin{aligned} cov(X_{t+h}, X_t) &= \sum_{j=1}^k cov(A_j c_{t+h,j}, A_j c_{t,j}) + cov(B_j s_{t+h,j}, B_j s_{t,j}) \\ &= \sum_{j=1}^k \sigma_j^2 (c_{t+h,j} c_{t,j} + s_{t+h,j} s_{t,j}). \end{aligned}$$

Using the trigonometric identity $\cos(a)\cos(b) + \sin(a)\sin(b) = \cos(a-b)$, we simplify:

$$cov(X_{t+h}, X_t) = \sum_{j=1}^k \sigma_j^2 c_{h,j} = \gamma(h).$$

- X_t is stationary since its mean and auto-covariance are independent of t . Then the spectral density is defined as the Fourier transform of the auto-covariance function:

$$f(\omega) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \omega h} = \sum_{h=-\infty}^{\infty} \left(\sum_{j=1}^k \sigma_j^2 c_{h,j} \right) e^{-2\pi i \omega h}.$$

Interchange the summations:

$$f(\omega) = \sum_{j=1}^k \sigma_j^2 \sum_{h=-\infty}^{\infty} c_{h,j} e^{-2\pi i \omega h}.$$

Using the trigonometric identity for cosine:

$$c_{h,j} = \frac{1}{2} (e^{2\pi i \omega_j h} + e^{-2\pi i \omega_j h}),$$

we substitute this into the inner summation:

$$\sum_{h=-\infty}^{\infty} c_{h,j} e^{-2\pi i \omega h} = \frac{1}{2} \sum_{h=-\infty}^{\infty} e^{i2\pi h(\omega_j - \omega)} + \frac{1}{2} \sum_{h=-\infty}^{\infty} e^{-i2\pi h(\omega_j + \omega)}.$$

These infinite summations result in the Dirac delta functions due to the orthogonality of complex exponentials:

$$\sum_{h=-\infty}^{\infty} e^{i2\pi h(\omega_j - \omega)} = \delta(\omega - \omega_j), \quad \sum_{h=-\infty}^{\infty} e^{-i2\pi h(\omega_j + \omega)} = \delta(\omega + \omega_j).$$

where the Dirac delta function is defined as

$$\delta(x) = \begin{cases} 1 & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Thus:

$$\sum_{h=-\infty}^{\infty} c_{h,j} e^{-2\pi i \omega h} = \frac{1}{2} (\delta(\omega - \omega_j) + \delta(\omega + \omega_j)).$$

Substituting this back into the spectral density:

$$f(\omega) = \sum_{j=1}^k \sigma_j^2 \cdot \frac{1}{2} [\delta(\omega - \omega_j) + \delta(\omega + \omega_j)].$$

Distribute the summation:

$$f(\omega) = \frac{1}{2} \sum_{j=1}^k \sigma_j^2 \delta(\omega - \omega_j) + \frac{1}{2} \sum_{j=1}^k \sigma_j^2 \delta(\omega + \omega_j).$$

Thus, the spectral density of the process is:

$$f(\omega) = \frac{1}{2} \sum_{j=1}^k \sigma_j^2 \delta(\omega - \omega_j) + \frac{1}{2} \sum_{j=1}^k \sigma_j^2 \delta(\omega + \omega_j).$$

4. Derive the periodogram of the following time series realization with n observations:

$$x_t = ac_{t,j} + bs_{t,j},$$

with a, b constants, ω_j the j -th Fourier frequency, $c_{t,j} = \cos(2\pi\omega_j t)$, and $s_{t,j} = \sin(2\pi\omega_j t)$.

Solution: The periodogram is defined as:

$$P(\omega_j) = |d(\omega_j)|^2 \text{ with } d(\omega_j) = \frac{1}{n} \sum_{t=0}^{n-1} x_t e^{-2\pi i \omega_j t}.$$

Substitute the time series realization $x_t = ac_{t,j} + bs_{t,j}$ into the Fourier transform:

$$d(\omega_j) = \frac{1}{n} \sum_{t=0}^{n-1} (ac_{t,j} + bs_{t,j}) e^{-2\pi i \omega_j t}.$$

Split the summation:

$$d(\omega_j) = \frac{a}{n} \sum_{t=0}^{n-1} c_{t,j} e^{-2\pi i \omega_j t} + \frac{b}{n} \sum_{t=0}^{n-1} s_{t,j} e^{-2\pi i \omega_j t}.$$

Consider the terms separately. Using the Euler formula $c_{t,j} = \frac{e^{2\pi i \omega_j t} + e^{-2\pi i \omega_j t}}{2}$:

$$\sum_{t=0}^{n-1} c_{t,j} e^{-2\pi i \omega_j t} = \frac{1}{2} \sum_{t=0}^{n-1} (e^0 + e^{-4\pi i \omega_j t}) = \frac{n}{2}.$$

due to the orthogonality of complex exponentials.

Similarly, and using the Euler formula $s_{t,j} = \frac{e^{2\pi i \omega_j t} - e^{-2\pi i \omega_j t}}{2i}$:

$$\sum_{t=0}^{n-1} s_{t,j} e^{-2\pi i \omega_j t} = \frac{1}{2i} \sum_{t=0}^{n-1} (e^0 - e^{-4\pi i \omega_j t}) = \frac{n}{2i}.$$

Substituting back into $d(\omega_j)$:

$$d(\omega_j) = \frac{1}{n} \left(a \cdot \frac{n}{2} + b \cdot \frac{n}{2i} \right).$$

Simplify:

$$d(\omega_j) = \frac{1}{2} (a - ib).$$

Thus, the periodogram evaluates to

$$P(\omega_j) = |d(\omega_j)|^2 = \left| \frac{1}{2} (a - ib) \right|^2 = \frac{1}{4} (a^2 + b^2).$$